Five course selections

Learn Web3

Web3 Fundamentals · 15 lessons

Wallets and Private Keys

Decentralized Finance · 9 lessons

Lending and Borrowing in DeFi

Smart Contract Development · 9 lessons

Testing Smart Contracts

Web3 Marketing & Community · 8 lessons

Growth Metrics and Analytics

AI x Crypto Intersection · 7 lessons

Autonomous Agents

Wallet control is not legal identity

Web3 Fundamentals · Lesson 5 of 15

In a self-custodial wallet, you control the keys or account permissions. Custodial services manage those controls on your behalf. A signature proves control of an account, not a person's legal identity.

Lesson summary

A wallet signs messages and transactions. A signature proves account control, not legal identity.

Borrowing cap and liquidation threshold differ

Decentralized Finance · Lesson 2 of 9 · Illustrative amounts

A collateralized position generally needs collateral worth more than the borrowed amount. The permitted loan-to-value ratio depends on the asset, market, and configured risk parameters.

Lesson summary

Example: 80% LTV on $1,000 collateral gives an $800 borrowing cap. Liquidation uses a separate threshold, with parameters set per market.

Turn a transfer into explicit assertions

Smart Contract Development · Lesson 5 of 9 · Foundry / Solidity

```
function test_Transfer() public {
token.transfer(alice, 100);
assertEq(token.balanceOf(alice), 100);
```

test_Transfer example

Sample raw units

The sample transfers 100 units from a 1,000-unit balance and asserts 100 received, 900 left. Foundry fuzz tests vary the amount to test properties.

Count wallets, then define the returning cohort

Web3 Marketing & Community · Lesson 5 of 8 · Measurement units

Define what each dataset measures: accounts, addresses, transactions, sessions, or people. Avoid combining these as though they were the same unit.

Lesson summary

Unique Active Wallets counts addresses over a defined period; one person can control several. Retention measures return activity from a starting cohort.

Reading a balance does not authorize a transfer

AI x Crypto Intersection · Lesson 2 of 7 · Autonomous Agents

Choose permissions per action. Reading a balance, drafting a transaction, and signing a transfer should not automatically receive the same level of access.

Lesson summary

Separate model, tools and wallet. Balance reads should not grant transfer-signing permission. Enforce spending caps and contract allowlists.